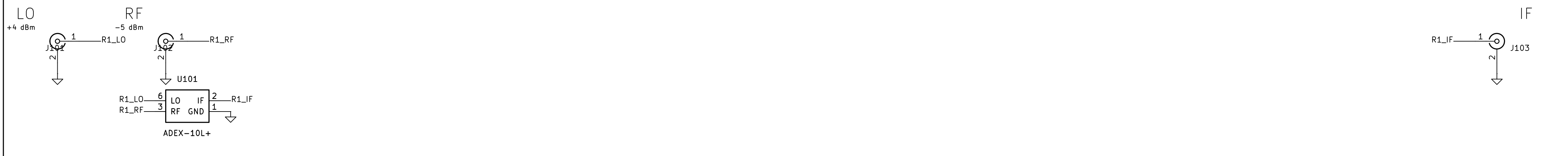
R1 diode mixer, ADEX-10L+

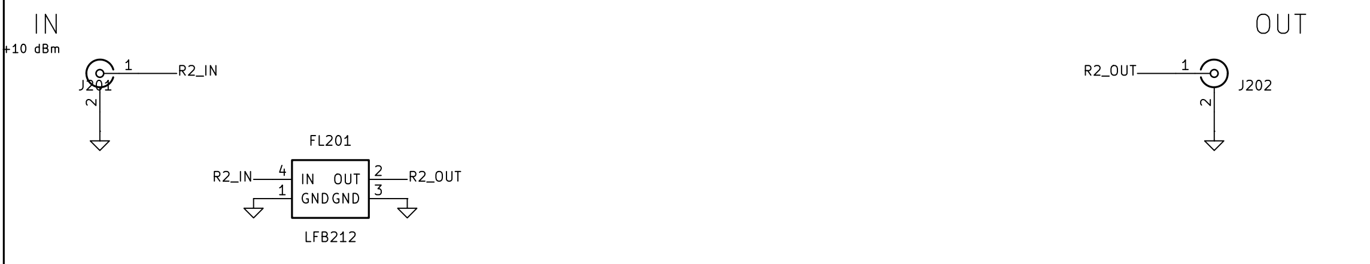
cells (0,0), (1,0)


LO
+4 dBm
RF
−5 dBm
IF
U101
ADEX-10L+
J103

EXPECT: about 7 dB of conversion loss from 10 MHz to 1 GHz with +4 dBm of LO (R10 at its top power), IF from DC to 800 MHz, 60 dB L–R isolation
Mini-Circuits ADEX-10L+: two wideband baluns and a ring of four Schottky diodes (drawn beside it on the silk). The LO (+4 dBm, +1 to +7) turns the diodes on in pairs, so the RF is multiplied by +−1 at the LO rate: the IF is RF x square wave, the sum and difference plus odd LO harmonics. Passive and DC-coupled at IF (a phase detector when RF and LO share a frequency); 1 dB compression at +1 dBm of RF, IP3 +16 dBm. Compare R7/R8 (ADL5350) and AMOD-A's A9 (MC1496).

R2 2.45 GHz filter

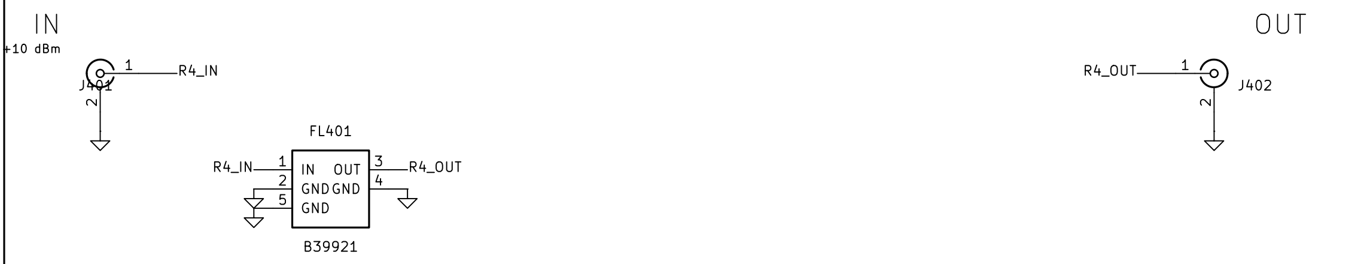
cell (2,0)


IN
+10 dBm
OUT
FL201
LFB212
J202

EXPECT: 1.7 dB of loss from 2.4 to 2.5 GHz, 30 dB or more below 2.2 GHz and above 2.7 GHz (the datasheet); 50 ohm in and out
Murata LFB212G45SG8A127, a 2012-size LTCC band-pass. The line to each jack is 50 ohm coplanar waveguide (0.29 mm wide, 0.25 to the ground pour) with a via fence; the jack's centre pin has no plane under it (a 2.2 mm void).
Sweep it with R10 or R13 into a power meter, or with a DAC harmonic comb and the ADC.

R4 915 MHz filter

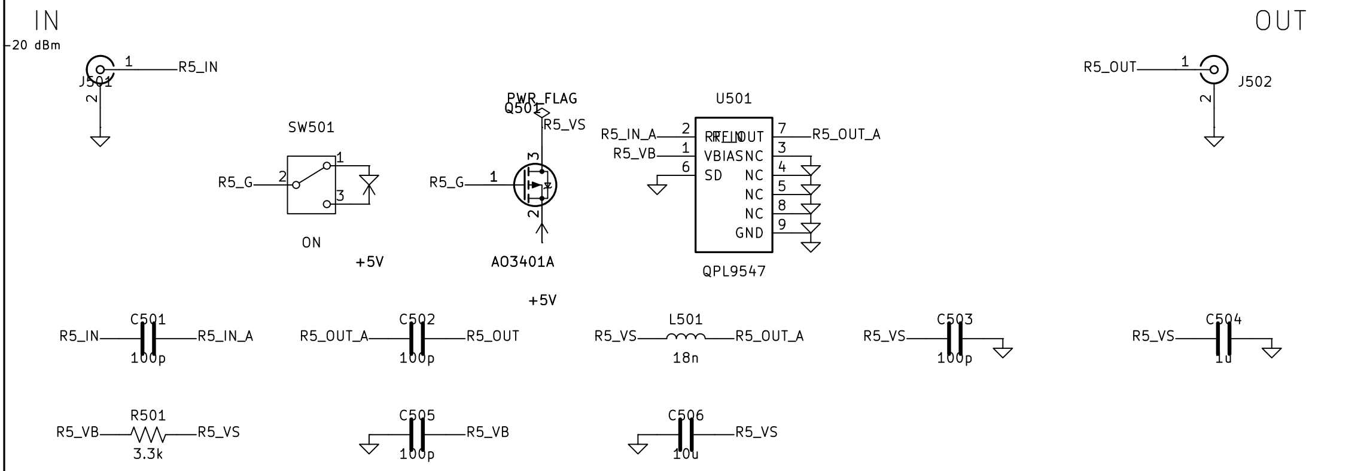
cell (3,0)


IN
+10 dBm
OUT
FL401
B39921
J402

EXPECT: 1.1 dB of loss from 902 to 928 MHz, 40 dB or more 30 MHz outside the band (the datasheet); 50 ohm in and out
RF360 B39921B2672P810: a surface-acoustic-wave filter for the 915 MHz ISM band, 26 MHz wide, in a 1411 package; its skirts are much steeper than the LTCC's. Lines, fences and launch voids as on R2.

R5 LNA, QPL9547

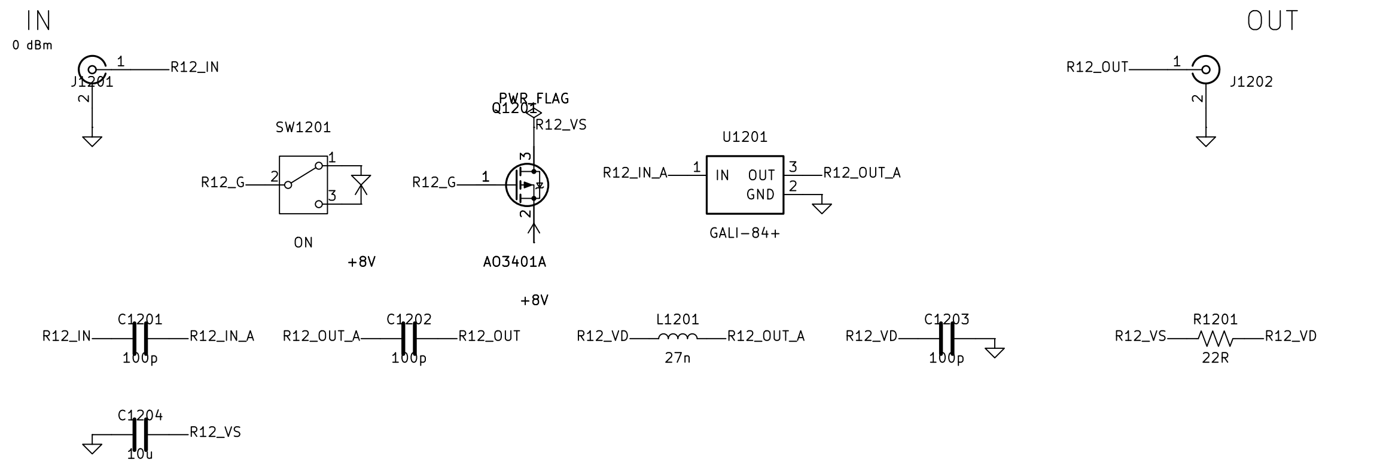
cell (0,1)


IN
−20 dBm
OUT
SW501
ON
AO3401A
U501
QPL9547
L501
18n
R501
3.3k
J502

EXPECT: 19.5 dB of gain and a 0.3–0.5 dB noise figure from 0.5 to 3 GHz (datasheet: 0.3 dB at 1.9 GHz, OIP3 +39 dBm); 65 mA from 5 V
Qorvo QPL9547: a GaAs pHEMT LNA, internally matched to 50 ohm, 0.1–6 GHz. It sets the noise figure of every receiver here, so put it first (before the filter) unless a strong neighbour overloads it; +39 dBm OIP3 makes that rare.
The 18 nH wire-wound choke (Qorvo's value) feeds 5 V into the output pin; 3.3 k from 5 V to Vbias sets 65 mA.
Never cable a transmitter into it without 30 dB of attenuation (22 dBm absolute maximum at the input).

R12 gain block, GALI-84+

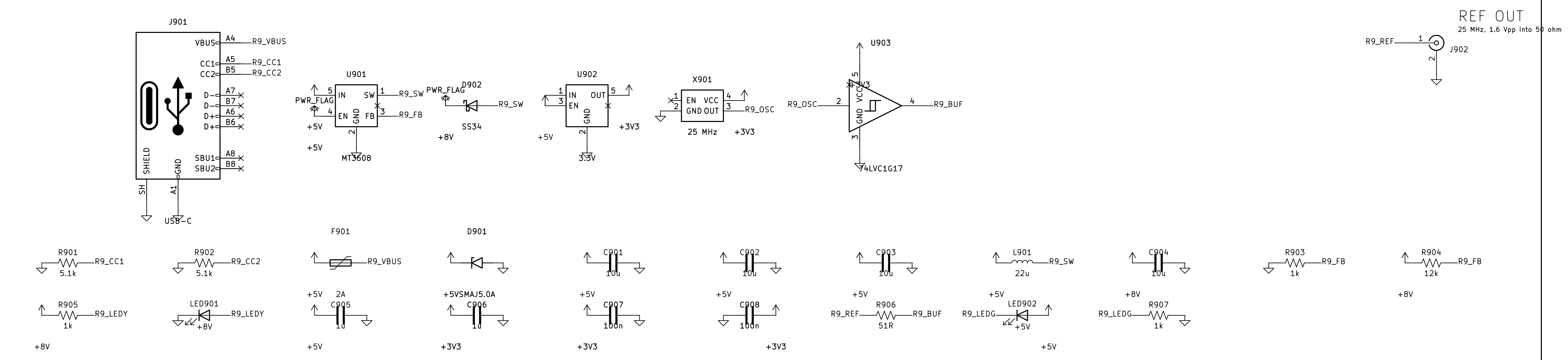
cell (1,1)


IN
0 dBm
OUT
SW1201
ON
AO3401A
U1201
GALI-84+
L1201
27n
R1201
22R
J1202

EXPECT: 20 dB of gain from 50 MHz to 2.5 GHz (datasheet: 21.9 dB at 1 GHz, 1 dB compression at +20 dBm out); 97 mA from 7.8 V
Mini-Circuits GALI-84+: a Darlington MMIC that sets its own current; the 22 ohm from the 7.8 V rail drops to its 5.8 V device voltage (0.2 W in the 2512). The 27 nH choke (415 ohm at 2.45 GHz, 155 at 915 MHz) feeds the output pin; 100 pF blocks keep the DC off the jacks. Keep the input under 0 dBm for linear gain. The slide switch and P-FET gate the tile's 7.8 V as on AMOD-A.

R9 power, USB-C 5 V in: +5, +7.8 V; 25 MHz reference

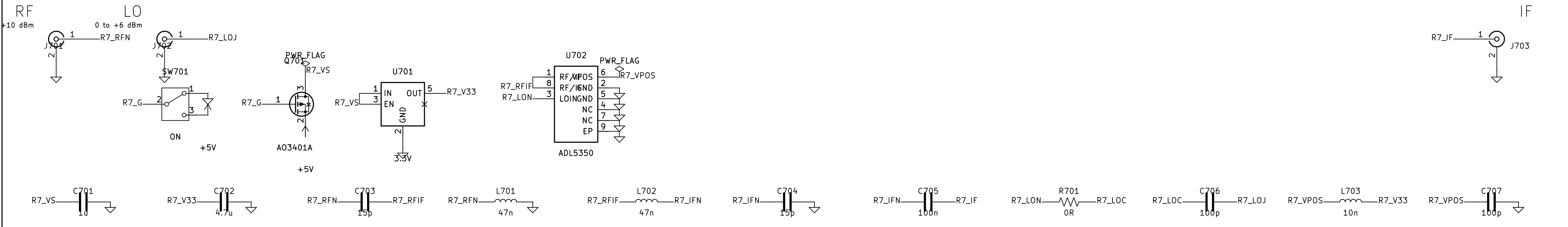
cells (2,1), (3,1)


REF OUT
25 MHz, 1.6 Vpp into 50 ohm
J901
USB-C
U901
MT3608
D902
SS34
U902
X901
25 MHz
U903
74LVC1G17
F901
2A
D901
SMAJ5.0A
LED901
LED902
J902
R901
5.1k
R902
5.1k
R905
1k
L901
22u

EXPECT: +5 V from the USB-C (up to 1.5 A), +7.8 V at 250 mA for the gain blocks (yellow LED), a 25 MHz square wave on REF OUT
USB-C power only: 5.1 k on each CC asks any source for 5 V; a 2 A PTC, a 5 V TVS and 20 uF. The MT3608 boosts to 7.8 V (0.6 V x 13 from 12 k / 1 k) for the GALI-84+ tile, R12 (100 mA); the +8V rail is a strip under row 1 that their headers reach with a via. The TPS7A2033 makes 3.3 V for the TC32H4 TCXO (+−2 ppm) and the 74LVC1G17 that drives REF OUT (51 ohm: 1.6 Vpp into 50 ohm, 3.3 Vpp into a synthesizer's unterminated REF IN; R10's REF IN passes it to R13).
The TCXO's pad 1 (enable or no connection on the drawing) floats, as CostasLink's review settled.

R7 TX mixer, ADL5350

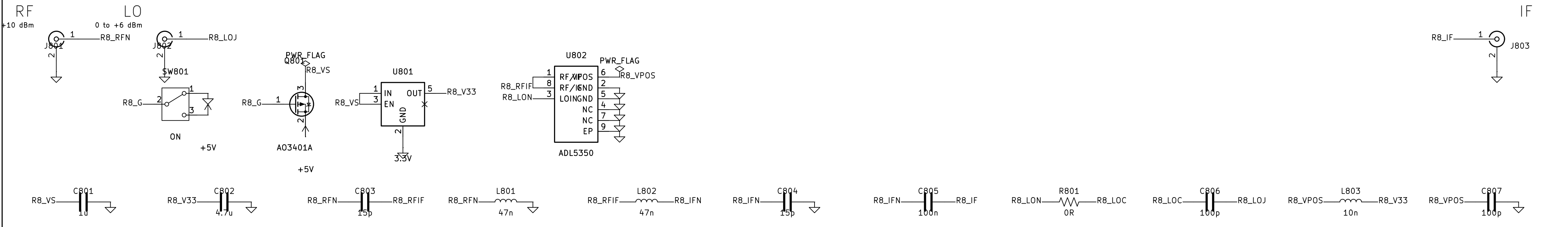
cells (0,2), (1,2)


RF
+10 dBm
LO
0 to +6 dBm
IF
SW701
ON
AO3401A
U701
U702
ADL5350
L701
47n
L702
47n
R701
0R
L703
10n
J703

EXPECT: RF x LO to IF (or IF x LO to RF) with 7–9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1–30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band-tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3–8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up-converts a DAC tone (IF in, RF out) or down-converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with R2 or R4.

R8 RX mixer, ADL5350

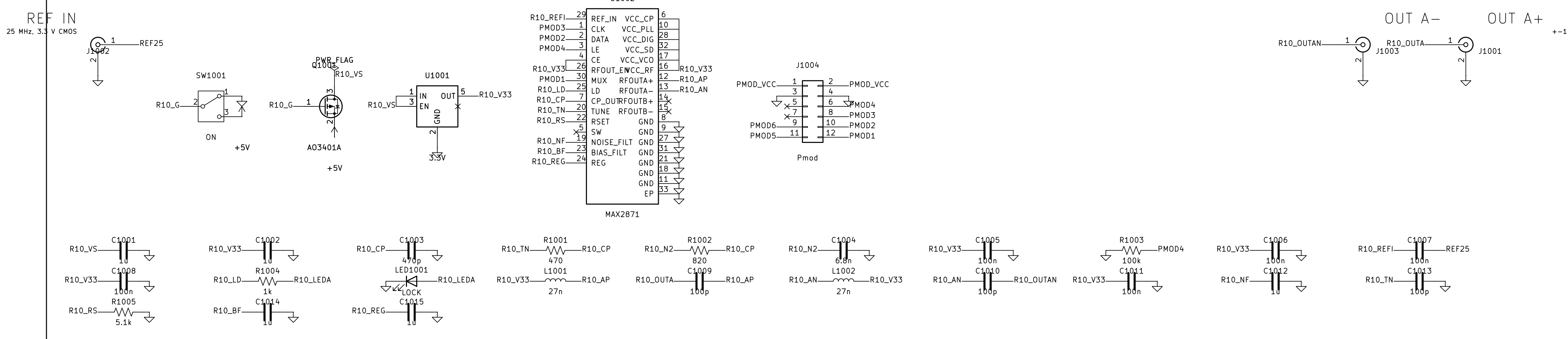
cells (2,2), (3,2)


RF
+10 dBm
LO
0 to +6 dBm
IF
SW801
ON
AO3401A
U801
U802
ADL5350
L801
47n
L802
47n
R801
0R
L803
10n
J803

EXPECT: RF x LO to IF (or IF x LO to RF) with 7–9 dB of loss from 400 MHz to 2.5 GHz and an IF of 0.1–30 MHz; +4 dBm of LO is enough
ADL5350: a GaAs FET switch mixer with its own LO buffer; the RF and IF pins are one node, split by a crossover (series 15 pF + shunt 47 nH toward RF, series 47 nH + shunt 15 pF toward IF, corner 200 MHz) in place of the datasheet's band-tuned tanks: a few dB more loss, every band at once. The LO's series element is 0 ohm (fit the datasheet's 3–8 nH for one band); the LO bias choke is 10 nH (the datasheet: 24 nH at 780 MHz, 2 nH at 2 GHz).
Passive, so the same tile up-converts a DAC tone (IF in, RF out) or down-converts an antenna's signal (RF in, IF out).
Both images appear: filter the RF side with R2 or R4.

R10 synthesizer, MAX2871

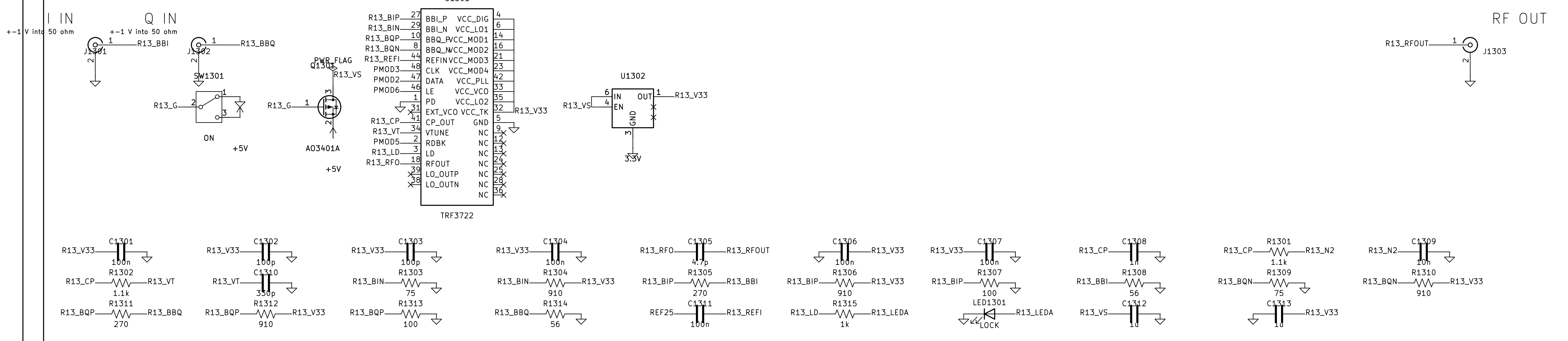
cells (0,3), (1,3)


REF IN
25 MHz, 3.3 V CMOS
OUT A−
OUT A+
SW1001
ON
AO3401A
U1001
U1002
MAX2871
J1004
Pmod
J1003
J1001
LED1001
LOCK
L1001
27n
L1002
27n

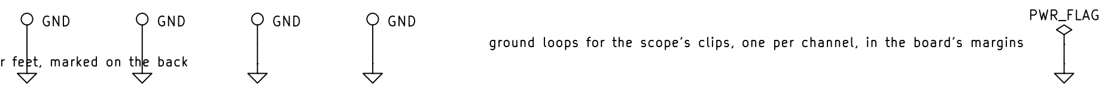
EXPECT: any frequency from 23.5 MHz to 6 GHz on OUT A+ and OUT A− (each −4 to +5 dBm in steps, 180 degrees apart), locked to the 25 MHz on REF IN; the red LED says locked
MAX2871 as CostasLink built it: fractional-N PLL with four VCOs, programmed over SPI from the Pmod (SCLK, SDATA, LE; MUXOUT back). The loop filter is third order, 80–120 kHz with 61 degrees of margin for both 915 MHz (VCO / 4) and 2.4 GHz (VCO / 2) at ICP 2.56 mA (RSET 5.1 k, CP = 7). RFOUTA+ and RFOUTA− each feed a jack through 27 nH to 3.3 V and 100 pF: two outputs, 180 degrees apart; put a 50 ohm terminator on one that is unused. RFOUTB is open and must stay disabled. CE and RFOUT_EN are tied high, LE has 100 k down. REF IN is unterminated; a back trace takes it on to R13.

R13 I/Q transmitter, TRF3722

cells (2,3), (3,3)


I IN
+−1 V into 50 ohm
Q IN
RF OUT
SW1301
ON
AO3401A
U1301
TRF3722
U1302
LED1301
LOCK
J1303

EXPECT: I + jQ from two DACs on 0.4–4.2 GHz, about −3 dBm, image-free (−42 dBc untrimmed)
TRF3722 (PLL, VCO, I/Q modulator) on SPI from the Pmod (LE IO6, readback IO5); reference from R10's REF IN. Each DAC: 56 ohm at the jack, 270 ohm to a node held at 0.25 V (+−0.25 V for +−1 V). DC on I alone is a CW carrier.


GND
GND
GND
GND
PWR_FLAG

no mounting holes: four stick-on rubber feet, marked on the back

ground loops for the scope's clips, one per channel, in the board's margins

Drawn by scripts/gen_sch.py from boards/rf_mod.py; the expected responses from sim/passive.py
Every tile is one circuit with its own SMA jacks; the board's silkscreen draws each tile's wires as they
a quadrature transmitter, a synthesizer, three mixers (one a diode ring), an LNA and a gain block, tw
Board: Jason Gallicchio and Claude | open hardware, CERN-OHL-P-2.0 | github.com/gallicchio/
Sheet: /
File: rf_mod.kicad_sch
Title: RF-MOD: 915 MHz, 2.4 GHz and a stretch to 5.8 GHz
Size: A1
Date: 2026-10-25
Rev: A
KiCad E.D.A. 10.0.7
Id: 1/1